

1. (Neural Network) Consider a two-layer network defined for a single training example:

$$\begin{aligned} z_1^{(i)} &= W_1 x^{(i)} + b_1, & a_1^{(i)} &= \text{ReLU}(z_1^{(i)}) \\ z_2^{(i)} &= W_2 a_1^{(i)} + b_2, & \hat{y}^{(i)} &= \sigma(z_2^{(i)}) \end{aligned}$$

The binary cross entropy loss is:

$$L^{(i)} = - \left(y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right), \quad J = \frac{1}{m} \sum_{i=1}^m L^{(i)}$$

Here:

- $x^{(i)} \in \mathbb{R}^{D_x \times 1}$ is the i -th input example
- $y^{(i)} \in \{0, 1\}$ is the corresponding label
- The hidden layer has D_{a_1} units, so $a_1^{(i)} \in \mathbb{R}^{D_{a_1} \times 1}$.

- (a) Shape of parameters

- (i) For a single example, give the shapes of:

$$W_1, b_1, W_2, b_2$$

- (ii) Now suppose we process all m training examples simultaneously (vectorized form).

Let:

$$X = [x^{(1)}, x^{(2)}, \dots, x^{(m)}], \quad Y = [y^{(1)}, y^{(2)}, \dots, y^{(m)}].$$

In vectorized form, we define

$$Z_1 = W_1 X + b_1, \quad A_1 = \text{ReLU}(Z_1), \quad Z_2 = W_2 A_1 + b_2, \quad \hat{Y} = \sigma(Z_2).$$

Give the shapes of:

$$X, Y, Z_1, A_1, Z_2, \hat{Y}.$$

- (b) For binary cross-entropy loss, the gradient with respect to the prediction is known to be

$$\frac{\partial J}{\partial \hat{Y}} = -\frac{1}{m} \left(\frac{Y}{\hat{Y}} - \frac{1-Y}{1-\hat{Y}} \right),$$

where

$$Y = [y^{(1)}, \dots, y^{(m)}], \quad \hat{Y} = [\hat{y}^{(1)}, \dots, \hat{y}^{(m)}].$$

Using the vectorized forward expression we defined in (a), derive the vectorized gradient

$$\frac{\partial J}{\partial W_2},$$

and write the corresponding gradient descent update rule for W_2 .

2. (Finite Hypothesis Class) Consider a binary classification problem with labels $y \in \{0, 1\}$. Let $\mathcal{H} = \{h_1, \dots, h_k\}$ be a finite hypothesis class with $|\mathcal{H}| = k$.

We draw a training set $S = \{(x_i, y_i)\}_{i=1}^n$ i.i.d. from a distribution $\mathcal{D}$.

For any hypothesis $h \in \mathcal{H}$, let

$$\hat{\varepsilon}(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{h(x_i) \neq y_i\},$$

be the **training error**, and

$$\varepsilon(h) = \Pr_{(x,y) \sim \mathcal{D}}(h(x) \neq y),$$

be the **generalization error**.

Let the ERM hypothesis be

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{\varepsilon}(h),$$

and let the best hypothesis in $\mathcal{H}$ be

$$h^* = \arg \min_{h \in \mathcal{H}} \varepsilon(h).$$

From the lecture, we know that for any $\gamma > 0$,

$$\Pr(\exists h \in \mathcal{H} : |\varepsilon(h) - \hat{\varepsilon}(h)| > \gamma) \leq 2k \exp(-2\gamma^2 n). \quad (\star)$$

(a) Using the bound $(\star)$, derive a condition on n (in terms of k , γ and δ) that guarantees

$$\Pr(\forall h \in \mathcal{H} : |\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma) \geq 1 - \delta.$$

(b) Use the bound $(\star)$ to show that, for any $0 < \delta < 1$, with probability at least $1 - \delta$ over the draw of the training set,

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + 2\sqrt{\frac{1}{2n} \log \frac{2k}{\delta}}.$$